SYNTHETIC DATA — 교육용 가상 데이터이며 실제 제품 사양·운영 기준과 무관합니다.

Online UPS 10kVA (UPS-LSU-10K) — Datasheet

1. Overview

Online double-conversion UPS 10kVA/9kW, single-phase. Product family: Uninterruptible power supplies. Lifecycle status: **MATURE**.

2. Technical specifications

2.1. Ratings

Item	Value
Rated power	10 kVA / 9 kW
Input / output	1-phase 220/230/240 V, 50/60 Hz
Battery	External, 16 x 12 V (192 V DC)

2.2. Performance and characteristics

Item	Value
Efficiency (online mode)	94%
Transfer time (mains to battery)	0 ms
Input voltage window (default)	176-276 V at full load

2.3. Endurance and communication

Item	Value
Communication	RS-232, USB, SNMP card slot

3. Installation data

3.1. Terminals

- Dimensions (W x H x D): 190 x 450 x 690 mm
- Weight: 28 kg
- Degree of protection: IP20
- Operating temperature: 0 to +40 °C

4. Standards and certifications

Applicable standards:

- IEC 62040-1
- IEC 62040-3

Certifications held:

- CE
- KC

Certificates and declarations of conformity are available from the Certification & Compliance team on request. Certifications not listed above are not held for this product; project-specific evaluation can be requested.

5. Ordering information

Item	Value
Ordering code	UPS-LSU-10K
Standard lead time	4 weeks ex-works
Minimum order quantity	1
Warranty	24 months (see POL-WARRANTY)
Firmware	latest 2.30 (see RN-UPS-LSU-10K)

6. Related documents

- IG-UPS-LSU-10K — Installation and Commissioning Guide
- TG-UPS-LSU-10K — Troubleshooting and Error Code Guide
- RN-UPS-LSU-10K — Firmware Release Notes
- FAQ-UPS — Uninterruptible power supplies FAQ

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Online UPS 10kVA (UPS-LSU-10K) — Installation and Commissioning Guide

1. Safety

Installation, wiring and commissioning must be performed by qualified personnel. Disconnect and lock out all power sources before work. Verify absence of voltage.

2. Receiving and storage

Inspect the packaging on arrival and report damage within 7 days (see POL-RETURN). Store indoors in a dry place between -25 and +55 °C, relative humidity below 85% without condensation.

3. Mounting and clearances

- Required clearances: 300 mm at the rear and 100 mm on each side for ventilation
- Operating temperature: 0 to +40 °C
- Degree of protection: IP20
- Above 1,000 m altitude, apply derating per the datasheet.

4. Wiring and tightening torques

- Terminal tightening torque: see dimension drawing
- Use copper conductors rated for at least 75 °C.
- Separate power and signal cables by at least 200 mm.

5. Commissioning procedure

1. Check battery polarity and voltage
2. Connect input and close input breaker
3. Run self-test
4. Perform battery test
5. Transfer load from bypass to inverter

6. Pre-energisation checklist

☐ Product rating matches the drawing
☐ All terminals tightened to specified torque
☐ Protective earth connected
☐ Insulation resistance measured and recorded
☐ Settings / parameters recorded

7. Common commissioning issues

Symptom	Likely cause	See
Short backup time	Battery end of life (> 3 years)	TG-UPS-LSU-10K §3.1
UPS on bypass due to overload	Connected load exceeds rating	TG-UPS-LSU-10K §3.2
Over-temperature alarm	Fan failure	TG-UPS-LSU-10K §3.4
SNMP card not detected	Firmware defect after hot-swap	TG-UPS-LSU-10K §3.5

On-site commissioning support can be requested through the Global CS Center. For uninterruptible power supplies the request is handled by the technical support team.

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Online UPS 10kVA (UPS-LSU-10K) — Firmware Release Notes

1. Summary

Latest firmware: **2.30**. Updating to the latest version is recommended for all units.

2. Version history

2.1. Version 2.30 (2025-11-17)

- Fixed: SNMP card not detected after hot-swap
- Fixed: battery test result not logged when load < 10%

2.2. Version 2.20 (2025-03-03)

- Added: generator mode with wider frequency window (45-65 Hz)

2.3. Version 2.10 (2024-07-15)

- Baseline release

3. Update procedure

1. Back up parameters / project before updating.
2. Connect the engineering tool (v4.6 or later).
3. Do not switch off power during the update.
4. After the update, verify the version and restore parameters if required.
5. If the update fails, do not power-cycle the unit repeatedly; contact support for the recovery procedure.

4. Compatibility

Parameter files from older versions can be loaded; files from newer versions cannot be loaded into older firmware.